

Security+ 3.4 · Backups

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.4** —
Backups (12:16) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Backup strategies involve frequency (hourly/daily/weekly) and retention (how many backups to keep). Incremental backups copy data changed since the last backup (any type). Differential backups copy data changed since the last full backup. Snapshots capture instantaneous state. Replication provides real-time copying. The 3-2-1 rule: 3 copies, 2 different media, 1 offsite.

High-Yield Core Concepts Tested on Exam

- Incremental: copies data since last backup (fast, small files)
- Differential: copies data since last full backup (slower, more robust)
- 3-2-1 rule: 3 copies, 2 media types, 1 offsite
- Offline backups survive ransomware attacks
- Snapshots: point-in-time copies, often stored locally
- Replication: real-time data copying to another location

Exam Tips & Classic Scenarios

- 'copies data since the last full backup' = differential
- '3 copies, 2 media types, 1 offsite' = 3-2-1 rule
- Incremental backups require the last full and all incrementals to restore

Typical Exam Scenario: A backup on Monday only copies files changed since Sunday's full backup. This is a differential backup.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.4 — Backups)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. An S3 bucket configured with public access block is protected against:

- A)** Unauthorized access via the internet
- B)** Accidental deletion
- C)** Weak encryption
- D)** Metadata leakage

2. Which S3 feature ensures 99.99999999% (11 nines) durability?

- A)** Encryption at rest
- B)** Versioning and cross-AZ replication
- C)** Lifecycle policies
- D)** Transfer Acceleration

3. Corporate scenario (Backups): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Incremental: since the last backup (fast, many small files).' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: Incremental: since the last backup (fast, many small files).

4. Corporate scenario (Backups): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Differential: since the FULL (less fast, more robust)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Differential: since the FULL (less fast, more robust)...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

5. Corporate scenario (Backups): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: '3-2-1: 3 copies, 2 media, 1 offsite; offline backup survives ransomware'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 3-2-1: 3 copies, 2 media, 1 offsite; offline backup survives ransomware...

6. Corporate scenario (Backups): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'copies data since the last full backup'=differential'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to speed up system throughput
- B)** Implement the corresponding solution: 'copies data since the last full backup'=differential...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

7. Corporate scenario (Backups): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: '3 copies, 2 media types, 1 offsite' = 3-2-1 rule. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: '3 copies, 2 media types, 1 offsite' = 3-2-1 rule
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Backups): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Monday's backup only copies what changed since Sunday's FULL' = differential'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'Monday's backup only copies what changed since Sunday's FULL' = differential...

9. Corporate scenario (Backups): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Incremental: since the last backup (fast, many small files).' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Incremental: since the last backup (fast, many small files).
- D)** Allow shared access among administrators using generic credentials

10. Corporate scenario (Backups): During a security review and compliance audit for the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Differential: since the FULL (less fast, more robust)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal adherence terms
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: Differential: since the FULL (less fast, more robust)...

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — Unauthorized access via the internet.

✓ **Why it's correct:** An S3 bucket configured with Block Public Access prevents bucket or object policies from granting unauthenticated public access via the internet. This means any anonymous or unauthorized access attempt from the internet is denied unless the requester has valid credentials and explicit permissions. This control is essential to prevent data leaks, which are a leading cause of cloud incidents. In CompTIA Security+, public access block configuration is an example of a cloud security control to protect the confidentiality of data stored in object storage services.

✗ **Why the others are wrong:**

- *Accidental deletion:* Accidental deletion is mitigated by controls such as versioning, MFA Delete, and retention policies (Object Lock). Public access block does not prevent accidental deletion because it only acts on access control, not on the integrity or availability of objects. Therefore, the alternative is wrong because it confuses a confidentiality control with an availability/integrity control.
- *Weak encryption:* Weak encryption refers to the use of inadequate algorithms or keys to protect data. Public access block has no relation to encryption strength; it only restricts who can access the bucket. Even with strong encryption, a public bucket can expose data if permissions are misconfigured. The alternative is incorrect because it addresses a different aspect of security (cryptographic strength) and not access control.
- *Metadata leakage:* S3 object metadata includes information such as name, size, modification date, and tags. Public access block prevents unauthorized access to content and also to metadata, but the main focus is content. Additionally, metadata leakage can occur through other means, such as logs or misconfigured APIs. The alternative is wrong because public access block is not specifically designed to protect metadata, but rather to deny public access to the bucket as a whole.

2) B — Versioning and cross-AZ replication.

✓ **Why it is correct:** The correct answer is 'Versioning and cross-AZ replication'. Amazon S3 provides 99.999999999% (11 nines) durability because it stores each object redundantly across multiple facilities within a region, distributed across at least three Availability Zones (AZs). Versioning protects against accidental deletions and overwrites by keeping previous versions of objects, while cross-AZ replication (and optionally cross-region replication) ensures data exists in physically separate locations. This combination of geographic redundancy and version history underpins the 11 nines durability. CompTIA addresses durability as the probability that data will not be lost, and in S3 this is achieved through automatic replication across AZs and versioning.

✗ **Why the others are wrong:**

- *Encryption at rest:* Encryption at rest protects data from unauthorized access when stored, using keys managed by S3 (SSE-S3), KMS (SSE-KMS), or the customer (SSE-C). It is a confidentiality measure, not durability. It does not prevent data loss from hardware failure, accidental deletion, or corruption; therefore, it is not the feature that guarantees 11 nines of durability.
- *Lifecycle policies:* Lifecycle policies are rules that automate the transition of objects between storage classes (e.g., Standard to Glacier) or expiration/deletion after a period. They help optimize costs and manage data lifecycle, but do not increase S3's intrinsic durability. In fact, misconfigured policies can lead to data deletion, reducing availability. It is not the mechanism that ensures 11 nines.
- *Transfer Acceleration:* Transfer Acceleration is a feature that uses CloudFront edge locations to accelerate long-distance uploads and downloads, improving transfer performance. It has no

relation to durability or storage redundancy; it is a network optimization. Therefore, it does not contribute to the 99.99999999% durability guarantee.

3) D — Implement the corresponding solution: Incremental: since the last backup (fast, many small files)..

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): Incremental: since the last backup (fast, many small files). Incremental backups only copy data that has changed since the last backup (whether full or incremental), making them fast and efficient for environments with many small files. The best recommendation is to implement an incremental backup strategy as part of a comprehensive backup plan.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

4) B — Implement the corresponding solution: Differential: since the FULL (less fast, more robust)....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): Differential: since the FULL (less fast, more robust). A differential backup captures all changes made since the last full backup. It is less fast to perform than an incremental backup (which only captures changes since the last backup of any type) but more robust because restoring requires only the last full backup and the latest differential backup, simplifying the restore process and reducing the risk of a corrupted chain. This is a data availability and recovery control.

✗ **Why the other options are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) D — Implement the corresponding solution: 3-2-1: 3 copies, 2 media, 1 offsite; offline backup survives ransomware....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): 3-2-1: 3 copies, 2 media, 1 offsite; offline backup survives ransomware. The 3-2-1 backup strategy ensures data resilience by maintaining three copies of data, on two different media types, with one copy stored offsite. An offline backup is critical because it is not accessible to ransomware, allowing recovery without paying a ransom.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: 'copies data since the last full backup'=differential....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): 'copies data since the last full backup'=differential.

✗ Why the others are wrong:

- *Temporarily disable audit logs to speed up system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) B — Implement the corresponding solution: '3 copies, 2 media types, 1 offsite' = 3-2-1 rule.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): '3 copies, 2 media types, 1 offsite' = 3-2-1 rule. This backup strategy ensures data redundancy and resilience by maintaining three copies of data, on two different media types, with one copy stored offsite.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
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8) D — Implement the corresponding solution: 'Monday's backup only copies what changed since Sunday's FULL' = differential....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): 'Monday's backup only copies what changed since Sunday's FULL' = differential. A differential backup copies all files that have changed since the last full backup. In the scenario, Sunday's full backup establishes the baseline; Monday's differential backup includes all

changes made since that full backup. On Tuesday, another differential backup would include all changes since Sunday's full backup (not just since Monday). This differs from an incremental backup, which only copies changes since the last backup of any type. Differential backups require more storage than incremental but simplify restoration because only the last full and last differential are needed.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

9) C — Implement the corresponding solution: Incremental: since the last backup (fast, many small files)..

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): Incremental: since the last backup (fast, many small files). Incremental backups only copy data that has changed since the last backup (whether full or incremental), making them fast and efficient for environments with many small files. The best recommendation is to implement an incremental backup strategy as part of a comprehensive backup plan.

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10) D — Implement the corresponding solution: Differential: since the FULL (less fast, more robust)....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Backups): Differential: since the FULL (less fast, more robust).

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accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.